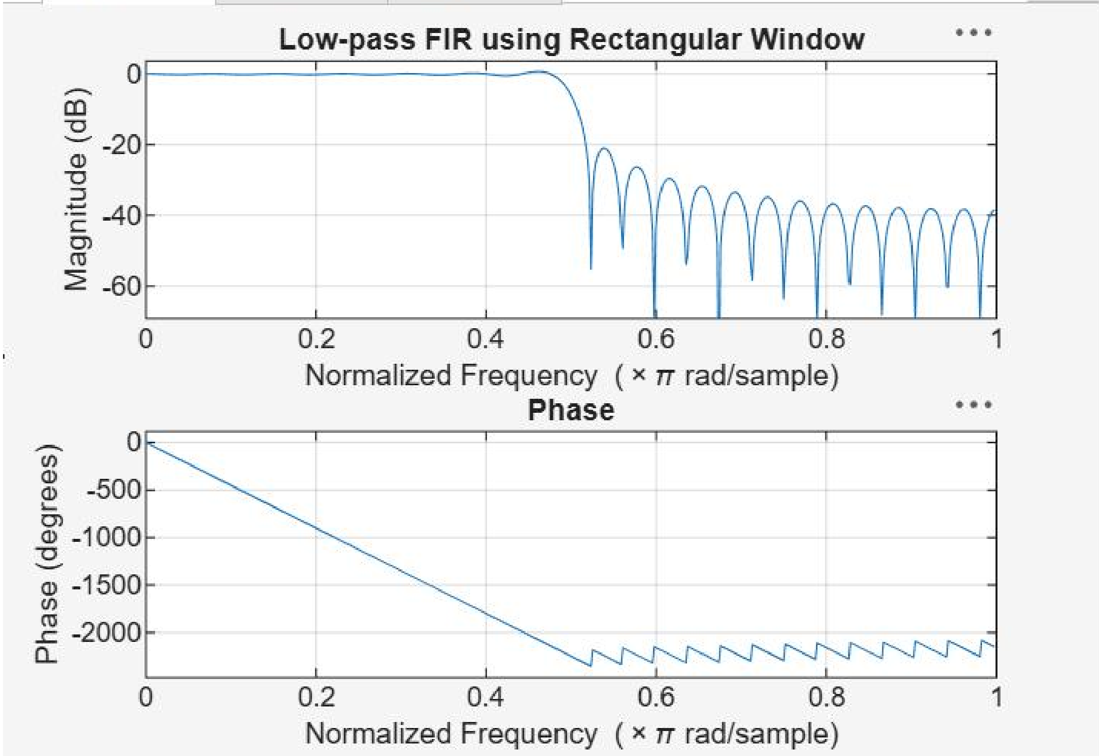
EXP 4 (4.1)

```
clc;
clear;
close all;
%% --- Input Parameters ---
fp = 1000; % Passband cutoff (Hz)
fs = 4000; % Sampling frequency (Hz)
N = 50; % Filter order (even number preferred)

wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)
%% --- 1. FIR Low Pass using Rectangular Window ---
b_rect = fir1(N, wc, 'low', rectwin(N+1));
disp('--- Rectangular Window Coefficients ---');
disp(b_rect');
%% --- 2. FIR Low Pass using Hamming Window ---
b_hamm = fir1(N, wc, 'low', hamming(N+1));
disp('--- Hamming Window Coefficients ---');
disp(b_hamm');
%% --- 3. FIR Low Pass using Hanning Window ---
b_hann = fir1(N, wc, 'low', hanning(N+1));
disp('--- Hanning Window Coefficients ---');
disp(b_hann');
%% --- Plot Frequency Responses ---
figure;
freqz(b_rect,1);
title('Low-pass FIR using Rectangular Window');
figure;
freqz(b_hamm,1);
title('Low-pass FIR using Hamming Window');
figure;
freqz(b_hann,1);
title('Low-pass FIR using Hanning Window');
```

```
--- Rectangular Window Coefficients ---
    0.0126
         0
   -0.0137
         0
    0.0150
         0
   -0.0166
         0
    0.0185
         0
   -0.0210
         0
    0.0242
         0
   -0.0286
         0
    0.0349
         0
   -0.0449
         0
    0.0629
```

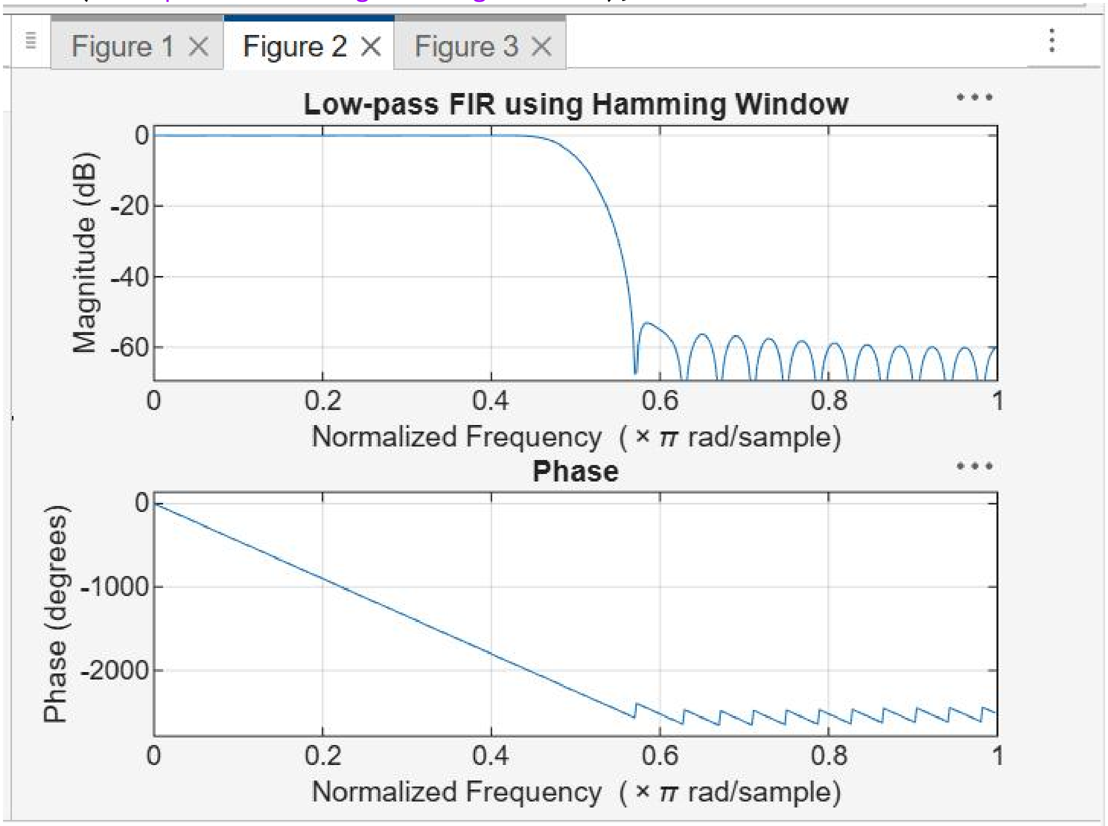
Figure 1 × Figure 2 × Figure 3 ×



EXP 4 (4.2)

```
clc;
clear;
close all;
%% --- Input Parameters ---
fp = 1000; % Passband cutoff (Hz)
fs = 4000; % Sampling frequency (Hz)
N = 50; % Filter order (even number preferred)
wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)
%% --- FIR Low Pass using Hamming Window ---

b_hamm = fir1(N, wc, 'low', hamming(N+1));
disp('--- Hamming Window Coefficients ---');
disp(b_hamm');
%% --- Plot Frequency Responses ---
figure;
freqz(b_hamm,1);
title('Low-pass FIR using Hamming Window');
```

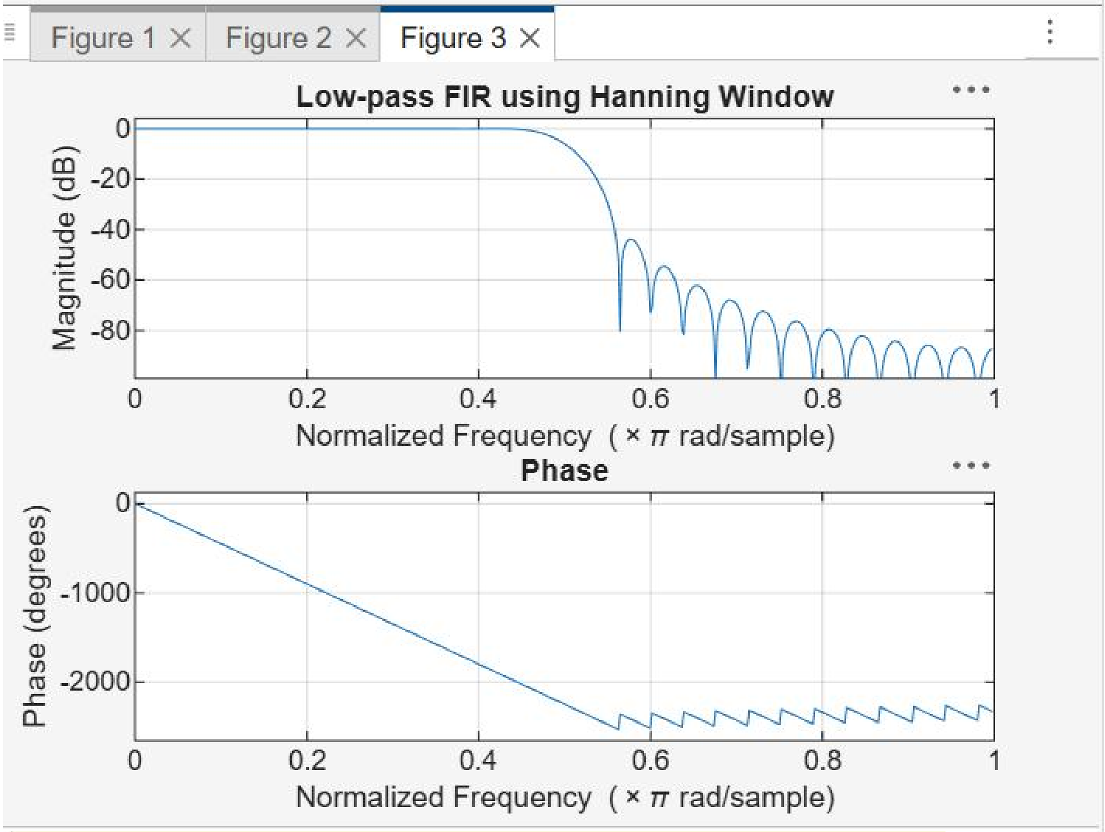


```
--- Hanning Window Coefficients ---
    0.0000
         0
   -0.0004
         0
    0.0013
         0
   -0.0028
         0
    0.0050
         0
   -0.0081
         0
    0.0122
         0
   -0.0179
         0
    0.0259
         0
   -0.0378
         0
    0.0580
```

EXP 4 (4.3)

```
clc;
clear;
close all;
%% --- Input Parameters ---
fp = 1000; % Passband cutoff (Hz)
fs = 4000; % Sampling frequency (Hz)
N = 50; % Filter order (even number preferred)
wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)

%% --- FIR Low Pass using Hanning Window ---
b_hann = fir1(N, wc, 'low', hanning(N+1));
disp('--- Hanning Window Coefficients ---');
disp(b_hann');
%% --- Plot Frequency Responses ---
figure;
freqz(b_hann,1);
title('Low-pass FIR using Hanning Window');
```




```
--- Hanning Window Coefficients ---
    0.0000
         0
   -0.0004
         0
    0.0013
         0
   -0.0028
         0
    0.0050
         0
   -0.0081
         0
    0.0122
         0
   -0.0179
         0
    0.0259
         0
   -0.0378
         0
    0.0580
```